SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	**Assessment:**	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA	**Total Marks:**	100
CO3 Statement:	*Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.*		
Student Name:	G.KOUSHIK	**Reg. No.:**	192524238
Section / Batch:	AIDS	**Date:**	10/08/2026

Code of Conduct

I koushik.g (192525452) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: Koushik.g

Instructions

- This test contains 25 MCQ Test questions (4 marks each), Total: 100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements: Identify the most appropriate answer, conceptual understanding, problem-solving ability, application of knowledge.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Preliminaries – Tree Terminology	Root, height, level, siblings and sub tree	1–2	8
Binary Tree, Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	3–9	28
Tree Traversals	Traversal types, In order, Post order and Pre order	10–14	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	15–16	8
Applications of Search Trees	Usage of binary tree	17	4
B Tree - 2-3 tree, 2-3-4 tree	Properties, Example and Operations	18–19	8
TRIE	Properties, structures and Operations	20	4
Red Black Tree	Properties, Balancing and Operations	21–22	8

Splay Tree	Splaying, Types and examples for insertion and deletion	23–25	12
	GRAND TOTAL:	**100 Marks**	

MCQ

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Application Skills	20	Correctly applies concepts to solve complex and scenario-based MCQs	Applies concepts correctly in most moderate-level questions	Struggles with application in complex scenarios	Unable to apply concepts effectively
Analytical Thinking	30	Consistently selects correct answers for high-order questions	Performs well in moderate analytical questions	Limited success in analytical questions	Unable to interpret analytical/problem -based questions
Time Management	20	Completes test efficiently with time for review	Completes test within time	Struggles to complete test on time	Unable to complete test
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____________	Signature: _____________	Signature: _____________
Date: ________________	Date: ________________	Date: ________________

SIMATS Online Programming Lab
DS PROGRAMMING
GANGINENI KOUSHIK
192524238RB

Back

Q1 **Tree Terminology** 4 marks

Nodes having the same parent are called:

A. Ancestors

B. Descendants

C. Siblings

D. Leaves

Save Answer | Next

SIMATS Online Programming Lab
DS PROGRAMMING
GANGINENI KOUSHIK
192524238RB

Back

Q2 **Quad Tree** 4 marks

In a Quad Tree, each internal node has at most:

A. 2 children

B. 3 children

C. 4 children

D. 8 children

Save Answer | Next

SIMATS Online Programming Lab

DS PROGRAMMING

GANGINENI KOUSHIK
192524238RB

Back

Q3 **Splay Tree** 4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf

B. Middle

C. Root

D. NULL

Save Answer Next

SIMATS Online Programming Lab

DS PROGRAMMING

GANGINENI KOUSHIK
192524238RB

Back

Q4 **Splay Trees** 4 marks

Identify the rotation:

50

\

70

\

90

A. Zag-Zag

B. Zig-Zig

C. Zig

D. AVL Rotation

Save Answer Next

Back

Q5 **Splay Tree**

4 marks

Splay Tree is a type of:

A. Complete Binary Tree

B. Self-adjusting Binary Search Tree

C. Heap Tree

D. Graph

Save Answer

Next →

Back

Q6 **Red-Black Tree**

4 marks

If a red node has a red child after insertion, what is this called?

A. Black-height problem

B. BST violation

C. Root violation

D. Red-Red violation

Save Answer

Next →

← Back

Q7 **Red-Black Tree** 4 marks

Insert sequence 70, 50, 90, 40 causes insertion of 40 under red parent. Uncle is red. What is needed?

A. Rotation

B. Recoloring

C. Deletion

D. No change

Save Answer Next →

SIMATS Online Programming Lab
DS PROGRAMMING
GANGINENI KOUSHIK
192524238RB

← Back

Q8 **Red-Black Tree** 4 marks

If black height differs in two paths from the same node, the tree becomes:

A. Valid

B. Invalid

C. Balanced

D. Complete

Save Answer Next →

Back

Q9 **Red-Black Tree**

4 marks

Every path from a node to its descendant NULL nodes must have the same number of

A. Black nodes

B. Red nodes

C. Leaf nodes

D. Parent nodes

Save Answer Next

SIMATS Online Programming Lab
DS PROGRAMMING
GANGINENI KOUSHIK
192524238RB

Back

Q10 **2-3 Tree**

4 marks

In a balanced 2-3 Tree, all leaves must be at:

A. Different heights

B. Root level

C. Random height

D. Same height

Save Answer Next

Q11 **2-3-4 Tree**

4 marks

Insert 10, 20, 30, 40 in a 2-3-4 Tree. What happens after inserting 40?

A. Tree becomes unbalanced

B. Node split may occur

C. Node is deleted

D. No change

Save Answer Next

SIMATS Online Programming Lab
DS PROGRAMMING

GANGINENI KOUSHIK
192524238RB

Back

Q12 **2-3 Tree**

4 marks

The maximum number of children in a 2-3 Tree node is:

A. 3

B. 2

C. 5

D. 4

Save Answer Next

SIMATS Online Programming Lab

DS PROGRAMMING

GANGINENI KOUSHIK
192524238RB

Back

Q13 **TRIE** 4 marks

In Trie, 'm' represents:

A. Number of nodes

B. Length of the key

C. Number of edges

D. Height of tree

Save Answer Next

SIMATS Online Programming Lab

DS PROGRAMMING

GANGINENI KOUSHIK
192524238RB

Back

Q14 **B-Tree** 4 marks

In a B-Tree of order 5, the maximum number of keys in a node is:

A. 4

B. 5

C. 6

D. 3

Save Answer Next

Back

Q16 **Search Tree**

4 marks

The main goal of using search trees is:

A. Increase complexity

B. Reduce storage to zero

C. Improve efficiency of data operations

D. Avoid using memory

Save Answer

Next

Q17 **AVL Trees** 4 marks

AVL trees are preferred over ordinary BST because they:

A. Use less memory always

B. Maintain balance for faster operations

C. Do not require insertion

D. Have no root node

Save Answer Next

SIMATS Online Programming Lab
DS PROGRAMMING
GANGINENI KOUSHIK
192524238RB
Back

Q20 **AVL Tree** 4 marks

The maximum height difference allowed between left and right subtrees is:

B. 1

C. 2

D. 3

Save Answer Next

SIMATS Online Programming Lab

DS PROGRAMMING

GANGINENI KOUSHIK
192524238RB

Back

Q21 **Binary Search Tree** 4 marks

A data compression system uses a tree-based model to assign shorter codes to frequently used patterns. You need a self-adjusting BST where frequency is not stored explicitly. Which is most appropriate?

A. AVL Tree

B. Threaded Binary Tree

C. B-Tree

D. Splay Tree

Save Answer Next

SIMATS Online Programming Lab

DS PROGRAMMING

GANGINENI KOUSHIK
192524238RB

Back

Q22 **Tree Traversal** 4 marks

Consider the binary tree:
A
/ \
B C
/ \ \
D E F
What is the Preorder traversal?

A. D B E A C F

B. A B D E C F

C. D E B F C A

D. B D E A C F

Save Answer Next

Back

Q23 **Tree Traversal**

4 marks

You are debugging a program that reconstructs the original root of a binary decision tree using traversal logs: Inorder and postorder. Given: Inorder: D B E A F C Postorder: D E B F C A What is the root?

A. A

B. B

C. C

D. D

Save Answer | Next

SIMATS Online Programming Lab
DS PROGRAMMING
GANGINENI KOUSHIK
192524238RB

Back

Q24 **Binary Tree**

4 marks

The maximum number of nodes in a binary tree of height h is:

A. h

B. 2^h – 1

C. 2^(h+1) – 1

D. h^2

Save Answer | Next

Q25 **Tree Terminology**

4 marks

A path from a node to the root is formed by its:

A. Descendants

B. Ancestors

C. Children

D. Leaves

Save Answer